42. DNSSEC and DNS Security Awareness

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Phase: Phase 4 — DNS
Estimated time: 1 day
Slug: `42-dns-dnssec`

Learning objectives

- Know what DNSSEC protects
- See AD flag in dig
- Understand it does not encrypt DNS
- Prefer DoH/DoT for privacy

Prerequisites

Complete prior lessons. Prefer a Linux VM for `ip`/`ss`/`nft` examples; macOS works for many `curl`/`dig` drills.

```
dig example.com +dnssec +noall +answer +comments
dig example.com +dnssec | grep -E 'flags:|RRSIG'
# DoH example via curl
curl -sH 'accept: application/dns-json' 'https://cloudflare-dns.com/dns-query?
name=example.com&type=A' | head -c 200; echo
```

DNSSEC → authenticity/integrity. DoH/DoT → privacy on path to resolver.

Common mistakes

- Thinking DNSSEC hides queries
- Broken middleboxes failing DNSSEC validation
- No monitoring of DNSSEC expirations for your zones

Exercises

Work in `exercises/42-dns-dnssec/`:

```
./test.sh
```

- **A:** make tests pass
- **B:** run 2–3 real commands from this lesson in your lab and note outputs
- **C:** rewrite answers from memory next day

Do not open `solutions/42-dns-dnssec/` until after an honest attempt.

Self-check

1. DNSSEC vs DoH?
2. dig flag to request DNSSEC records?
3. What problem DNSSEC does not solve?

Further reading

- https://www.rfc-editor.org/rfc/rfc4033